

Program : Diploma in Textile Technology	
Course Code : 6061B	Course Title: Statistical Quality Control in Textile Testing
Semester : 6	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To impart the method of construction frequency distribution , graphical representation of frequency distribution and different types of frequency curves
- To impart the knowledge of significance tests on Mean & SD
- To equip with various management and organization of quality control

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic Maths Knowledge		Mathematics I	1
Basic Maths Knowledge		Mathematics II	2
Fundamentals of Textile Testing		Textile testing	5

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive level
CO1	Construct Frequency distribution & Graphical representation of Frequency distribution.	15	Understanding
CO 2	Apply the concept of probability and significance tests on mean and SD	14	Applying
CO 3	Interpret quality control charts	15	Understanding
CO 4	Identify Management and organization techniques of Quality control	13	Understanding
	Series Test	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2		3					
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Construct Frequency distribution & Graphical representation of Frequency distribution.		
M1.01	Understand the Basic concepts of statistics	3	Understanding
M1.02	Understand elements graphical representation of data	4	Understanding
M1.03	Understand the measures of central Tendency and frequency curves	4	Understanding
M1.04	Understand the different measures of Dispersion	4	Understanding
Contents: Definition of statistics - Difference between Quality control & Quality Assurance - Meaning of population and individual in statistics - Classification and tabulation of data - Preparation of continuous frequency distribution - Different types of Graphical representation of frequency distribution - Histogram & Frequency polygon - Different types of frequency curves - Normal distribution curve, Asymmetric distribution curve, Bi-modal distribution curve - Measures of central Tendency - Mean, Median, Mode - Measures of Dispersion - Range, Mean range, Mean deviation, Percentage mean deviation, Standard deviation & Co-efficient of variation - Characteristics of normal distribution curve - Relationship between mean range, mean deviation and SD.			
CO2	Apply the concept of probability and significance tests on mean and SD		
M2.01	Understand the Probability theory	5	Understanding
M2.02	Understand the Standard error	5	Understanding
M2.03	Familiarize the Significance Test	4	Understanding
	Series Test - I	1.5	
Contents:			

Probability theory - Sampling versus hundred percent inspections - Standard error - Single mean of small sample, Single mean of large sample, Between means of small sample, Between means of large sample - Significance Test - Significance test on Mean - Significance test on SD - Single SD of small sample, Single SD of large sample, Between SD's of small sample, Between SD's of large sample

CO3	Interpret quality control charts.		
M3.01	Understand the Concept of quality control charts	3	Understanding
M3.02	Understand the Fraction defective chart (p Chart)	4	Understanding
M3.03	Understand the Number defective chart (c Chart)	4	Understanding
M3.04	Understand the Construction of control chart	4	Understanding

Contents:

Concept of quality control charts - Meaning of control charts - Uses of quality control charts -Types of quality control charts - X and R chart - Fraction defective chart (p Chart), Number defective chart (c Chart) - Determination of control limits - Construction of control chart - Interpretation of control chart

CO4	Identify Management and organization techniques of Quality control		
M4.01	Understand the management techniques for quality control.	3	Understanding
M4.02	Understand Snap survey techniques	3	Understanding
M4.03	Understand the application of quality standards in Textile Mills	3	Understanding
M4.04	Understand the planning and organization of Total Quality Control in textile mills	4	Understanding
	Series Test - II	1.5	

Contents:

Management techniques for quality control - Statistical concepts and its tolerances –Machine interference with special reference to textile industry - Snap survey techniques - SQC systems adopted in Textile mills - Total Quality control - Planning and organization of Total Quality Control in mills - Cost reduction and quality control.

Text / Reference:

T/R	Book Title/Author
T1	J. E. Booth- Principles of Textile testing- Mahajan publishers Ahammedbad
T1	B.Grover and D.S.Hamby -Hand Book of Textile Test and QC
R1	T. Murphy, Tipper & KM. Morris -Statistical Methods for Textile Technologists
R2	S.P. Gupta -Statistical Methods -Universal Publishing corp.,Mumbai -2
R3	SITRA Publication -Methods of Statistics -Universal Publishing corp.,Mumbai -2

Online resources:

Sl.No	Website Link
1	http://nptel.ac.in/
2	www.textilelearner.blogspot.com
3	http://mytextilenotes.blogspot.in
4	http:// www.sitra.org.in
5	http://www.nitra.org.in